

# **Academic Paper Analysis & Critique - Michael Alpin**

## **Student Name**

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## **Chosen Cyber Technique**

MitM Evil Twin attack

## **Target Paper**

J. M. Chauhan, "Optimizing Rogue Access Point Detection with CART and Deep Learning Techniques," M.Sc. research project, National College of Ireland, Dublin, Ireland, 2023.

## **Paper Meta-Information & Research Group Focus**

### **Paper author**

Jay Chauhan

### **Publication Venue & Affiliation**

School of Computing at the National College of Ireland

### **Funding/Sponsorship**

No sponsors & self funded (with the university)

# Problem Statement & Literature Gap Analysis

## Previous work

The paper mentions the previous work done regarding the detection of the evil twin attack - a.k.a detection of rouge access points and network intrusion detection.

Here are the past papers and information:

| Research Paper            | Algorithms                                                | Limitations                                                                                                            | Upsides                                                                                 |
|---------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Kim, D. et al. (2018)     | SVM, Decision Tree, KNN, MLP                              | Narrow focus on RTT values, may not represent all unauthorized AP characteristics                                      | Effective comparison of ML techniques, KNN shows highest accuracy                       |
| Shetty, S. et al. (2007)  | Traffic analysis                                          | Reliance on specific traffic patterns and thresholds, risk of false positives/negatives                                | Automated, adaptable to various networks, independent of specific wireless technologies |
| Wu, W. et al. (2018)      | k-medoid algorithm for clustering analysis on RSS vectors | Focus on RSS data, challenges with complex datasets                                                                    | Reducing false alarm rate & high detection rate                                         |
| Vaidya, T. S. (2023)      | KNN, SVM, Random Forest, Genetic Algorithm                | Not identifying the type of unauthorized AP attack, only detects the simple pattern by using ML, Potential overfitting | Random forest shows the highest accuracy                                                |
| Baby, B. S. et al. (2023) | Random Forest, Decision Tree, SVM                         | Dataset specificity, algorithmic limitations, evaluation constraints                                                   | High accuracy (97.38%) and low error rate (0.25%)                                       |
| Hossain, Z. et al. (2021) | Decision Tree, Random Forest, SVM, Naïve Bayes            | Binary classification, lack of feature selection,                                                                      | Improved performance with popular classifiers                                           |

|                               |                                                                     |                                                                           |                                                             |
|-------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------|
|                               |                                                                     | dataset specificity                                                       |                                                             |
| Dsouza, A. et al. (2022)      | Naïve Bayes, SVM, Random Forest, Decision Tree, Logistic Regression | Dataset limitations, realtime processing challenges, model generalization | Feasible approach for realtime detection with high accuracy |
| Shone, N. et al. (2018)       | Nonsymmetric Deep AutoEncoder (NDAE) and Random Forest              | Potential overfitting, old dataset                                        | Improved detection accuracy, promising for modern NIDS      |
| Wei, P. et al. (2019)         | Deep Belief Network (DBN) with optimization algorithms              | Performance overhead, real life application challenges                    | Promising classification accuracy                           |
| Alsughayyir, B. et al. (2019) | Deep Autoencoders for feature learning and classification           | Potentially inaccurate on unseen data                                     | High accuracy                                               |

## Identified limitations

As we can see in the table, different techniques and algorithms were previously used to implement the detection. Though those attempts have several gaps & flows:

- False positives / negatives
- Focused on specific types & Potential overfitting
- Performance overhead
- Old datasets & Not affective on zero-day variants

## The Core Research Question

The main focus of this paper is to develop an advanced and efficient method for detecting rogue access points in wireless networks, while considering the limitations of traditional machine learning models, addressing the key challenges observed in previous studies and enabling the model to be more adaptable and effective against contemporary network threats.

## Model Selection & Architectural Justification

The study used 3 models for the development: CART, FCNN and Ensemble model:

| Selected Model / Architecture             | Core Technical Mechanism                                                                                                                                                                                            | Reason for Selection (Why this model?)                                                                                                                                                                                                          |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CART (Classification And Regression Tree) | Uses a binary tree structure to divide data based on features where every node is a decision point                                                                                                                  | It has exceptional interpretability & It's comparatively cheap in computational cost                                                                                                                                                            |
| FCNN (Fully Connected Neural Network)     | Uses 4 layers (input, 2 hidden layers with 128 neurons each, and output). Extracts features from the data and processes them with the neural network layers with the ReLU & Sigmoid (in the output layer) functions | It has great capacity to identify the complicated connections and patterns within the dataset & Its layered structure helps to interpret the nuanced network traffic                                                                            |
| Ensemble model (FCNN + XGBoost)           | Combines both model predictions. XGBoost builds decision trees sequentially with each tree attempting to correct the mistakes made by the previous one                                                              | It takes advantage of both machine learning and deep learning approaches. XGBoost has high performance and accuracy in analyzing the structured data, and FCNN shows excellent performance in detecting the detailed, non-linear trends in data |

The author did not compare their chosen architectures against other baseline models.

# Hyperparameters & Model Optimization

The author mentions a few of the hyperparameters of the MLs in the article.

The rest of the hyperparameters he used that are listed were deduced from the source code listed in the configuration manual of the paper.

| Hyperparameter Category      | Specific Value/Setting                                                                                                                                                                      | Technical Justification / Selection Method                                                                                                                                                        |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Depth/Width [FCNN]   | 4 layers: <ul style="list-style-type: none"><li>• Input (12 features)</li><li>• #1 hidden layer (128 neurons)</li><li>• #2 hidden layer (128 neurons)</li><li>• Output (1 neuron)</li></ul> | The layered structure helps to interpret the nuanced network traffic                                                                                                                              |
| Activation Function [FCNN]   | <ul style="list-style-type: none"><li>• Hidden layers: Rectified Linear Unit (ReLU) function</li><li>• Output layer: Sigmoid function</li></ul>                                             | The ReLU activation function includes non-linearity, which is needed to model the complex relationship, and the Sigmoid activation function is very useful for binary classification applications |
| Optimization and Loss [FCNN] | Adam optimizer and Binary Crossentropy loss                                                                                                                                                 | Useful for binary classification problem                                                                                                                                                          |
| Learning Rate [FCNN]         | 0.001 with Adam Optimizer                                                                                                                                                                   | -                                                                                                                                                                                                 |
| Batch Size & Epochs [FCNN]   | Batch: 256<br>Epochs: 30                                                                                                                                                                    | -                                                                                                                                                                                                 |
| Regularization [FCNN]        | Dropout rate = 0.5                                                                                                                                                                          | -                                                                                                                                                                                                 |
| Max Depth [CART, XGBoost]    | None                                                                                                                                                                                        | -                                                                                                                                                                                                 |

# Data Pipeline & Implementation Workflow

## Data Preprocessing & Cleaning

- From the range of the attack files, focus was made on 4 specific types of network attacks (one of them being Evil Twin). Other attack data were removed from the dataset, so that research critically concentrate on the most impactful security issues
- The CSV files (taken from the AWID 3 dataset) were cleaned of any irrelevant and null values.
- After cleanup, the data pertaining to each attack type was merged into a single CSV file
- The network traffic data were encoded into binary format
- The normal traffic was labeled as '0', while the attack data was labeled as '1'

## Feature Engineering / Selection

- The ANOVA (Analysis of Variance) test played the significant role in the feature selection process
- Using the ANOVA test, the features with the highest variance value were identified
- Using the results, the dataset was then reduced to the 12 features that has impact on the predictability of the traffic data, as well as an extra label variable, indicating whether the traffic is normal or attack
- Finally, the amount of columns in the dataset was reduced from 34 to 13 (The author did not list the features tested & used)

## Software Stack & Frameworks

Python was used throughout the study.

The following libraries & tools were used:

- Pandas - for data manipulation
- Numpy - for numerical computations
- Scikit-learn - for implementing the CART model and data preprocessing tools
- TensorFlow & Keras - for training the deep learning FCNN model
- XGBoost - for developing the XGBoost model
- Matplotlib & Seaborn - for visualizing the results

## Experimental Environment & Quantitative Results

### Benchmark Datasets Used

The AWID 3 dataset was used for the following reasons:

- Its comprehensive array of network traffic data
- It is particularly relevant to contemporary network security challenges
- It has a variety of network behaviors, capturing both standard operational data and multiple attack scenario, which provides the realistic environment for testing and training the detection model

## Evaluation Metrics Defined

For the evaluation, the author used cross-validation.

In cross-validation, the whole dataset is being divided into K parts.

The first K-1 datasets are used for learning, and the last dataset is used for testing & evaluation. This learning & testing process is then being repeated K times, where each time the testing dataset is different, and the result is being calculated based on the average performance of the K runs.

The author used the following metrics for evaluation:

- Confusion Matrix - Table containing the rates of True Positives , True Negatives, False Positives and False Negatives (for example: the False Positive rate is the percentage of normal data that the model has predicted to actually be attack data).
- Precision and Recall - Precision is the percentage of correct positive prediction and Recall is the real positive prediction percentage
- F1 Score - harmonic mean of precision and recall

The results from the confusion matrix and cross validation are used to compare the models against each other.

## Reported Detection Performance

The paper does not provide comparison to a baseline model, so I show instead a comparison to the other models tested by the author.

| Cyber Technique Model Variant   | Accuracy | Precision | Recall (TPR) | False Positive Rate (FPR) |
|---------------------------------|----------|-----------|--------------|---------------------------|
| [Best Model] CART               | 99.97%   | 98.44%    | 98.56%       | 0.0016%                   |
| FCNN                            | 68.21%   | 33.98%    | 50.58%       | 21.44%                    |
| Ensemble model (FCNN + XGBoost) | 81.98%   | 88.11%    | 73.96%       | 1.799%                    |

## **Novelty & Scientific Contribution**

### **The "Secret Sauce"**

In the paper, the author is testing different and diverse model approaches to the problem, while keeping in mind the need for real time use, the need for good performance on completely new data, and the need for high accuracy, and low false positive rates. All of those principles that the author has kept in mind while testing for the best solution make this paper significant, and the results reliable.

### **Theoretical vs. Practical Value**

The CART model's implementation makes its computational cost comparatively cheap, making it a good choice when speedy response time is essential. On top of that, different optimization techniques like pruning (which the author uses) can significantly improve the performance, making it practical for real life application.

## **Actionable Engineering Takeaways for My Project**

As the result shown is very promising, here are my biggest takeaways from the paper:

- Consider using the CART model, or a similar technique
- Use the cross-validation & confusion matrix measurements
- Use Pandas, Numpy, Scikit-learn, TensorFlow & Keras, Matplotlib and Seaborn
- Make sure to avoid potential overfitting by making tests with various datasets
- Use ANOVA to find fitting features to train on